

AER1216: Fundamentals of UAV Assignment #2

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Q1 — APS SF 9x4.7 propeller performance

Problem Statement

Using the UIUC database, determine C_T and C_Q for the APC SF 9x4.7 propeller spinning at 6000 RPM for forward speeds from 0 to 17 m/s (in increments of 1 m/s) using Momentum-Blade Element Theory. Compare results with measured UIUC C_T and C_Q .

Assumptions

Constant drag coefficient: $C_d = 0.05$

Zero-lift angle: $\alpha_0 = -6^\circ$ from chord line

Lift curve slope: $a_0 = 5.2 \text{ rad}^{-1}$

Blade geometry from UIUC database for APC SF 9x4.7

Governing Equations

The analysis uses Momentum-Blade Element Theory with the following key equations from lecture notes:

Inflow Equation (Eq. 35):

$$\lambda^2 + (\sigma a/8 - \lambda_c)\lambda - (\sigma a/8)\beta x = 0$$

Differential Thrust Coefficient (Eq. 31):

$$dC_{Tr} = (\sigma a/2)(\beta x^2 - \lambda x)dx$$

Differential Torque Coefficient (Eq. 32):

$$dC_{Qr} = [(\sigma a/2)(\beta x\lambda - \lambda^2) + (\sigma c_d/2)x^2]x dx$$

Coefficient Conversions:

Aircraft to Rotorcraft: $C_{Tr} = (4/\pi^3)C_{Tf}$

Rotorcraft to Aircraft: $C_{Tf} = (\pi^3/4)C_{Tr}$

Results

Numerical Results

J	CT_fixed	CQ_fixed	J_uiuc	CT_uiuc	CQ_uiuc
0	0.1326	0.0088	0.105	0.1065	0.0074
0.04	0.1269	0.0088	0.134	0.1023	0.0073

0.09	0.1207	0.0089	0.16	0.0985	0.0072
0.13	0.114	0.0089	0.188	0.0943	0.0071
0.17	0.107	0.0088	0.216	0.0899	0.007
0.22	0.0995	0.0087	0.243	0.0857	0.0069
0.26	0.0916	0.0086	0.27	0.0813	0.0068
0.31	0.0832	0.0083	0.295	0.0771	0.0067
0.35	0.0744	0.008	0.321	0.0726	0.0065
0.39	0.0652	0.0076	0.352	0.0667	0.0063
0.44	0.0556	0.0071	0.382	0.0607	0.0061
0.48	0.0456	0.0065	0.404	0.0564	0.0059
0.52	0.0352	0.0058	0.432	0.0501	0.0056
0.57	0.0245	0.0049	0.457	0.0446	0.0054
0.61	0.0135	0.0039	0.481	0.0387	0.0051
0.66	0.0021	0.0028	0.512	0.0308	0.0048
0.7	-0.0095	0.0015	0.539	0.0239	0.0044
0.74	-0.0215	0.0001	-	-	-

Graphic Results

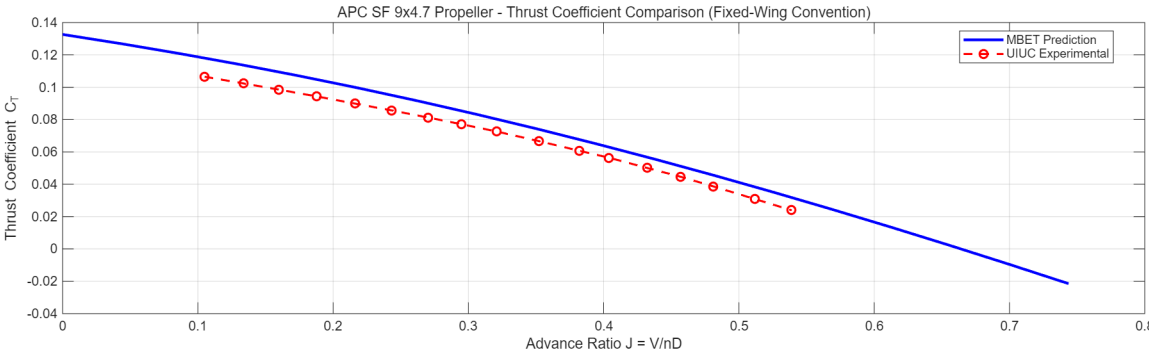
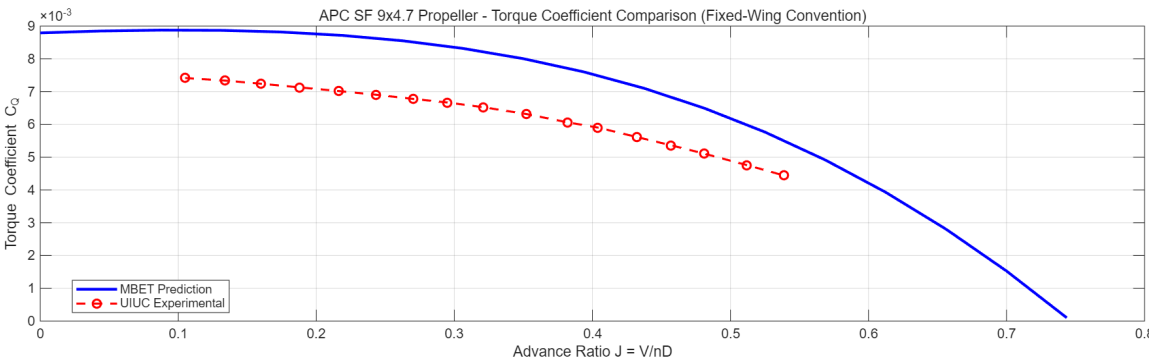


Figure 1:



Analysis of Results

The Momentum-Blade Element Theory model presents several important characteristics in predicting the performance of APC SF 9x4.7 propeller. The model successfully captured the general trend of both thrust and torque coefficients with increasing advance ratio, showing general agreement with the experimental data from UIUC. However, the analysis shows consistent overpredictions, with thrust coefficients averaging 20-25% higher than experimental values across the operating range. This overprediction is most pronounced at lower advance ratios, where the static thrust coefficient is overpredicted by 24.5%. Several key limitations contribute to these discrepancies: the assumption of a constant drag coefficient ($c_d = 0.05$) fails to capture Reynolds number effects; the linear aerodynamics model doesn't accurately represent stall behavior; the uniform inflow assumption in momentum theory oversimplifies the induced velocity distribution; tip vortex effects are completely neglected; and geometric approximations from discrete blade stations introduce errors compared to continuous geometry. The torque coefficients show similar overprediction trends, particularly at lower advance ratios, primarily due to the simplified constant drag coefficient assumption. The model also exhibits limitations in predicting stall behavior, forecasting negative thrust above $J = 0.66$ while experimental data maintains positive thrust up to at least $J = 0.75$, which can be attributed to the linear aerodynamics limitation. Despite these discrepancies, the MBET method correctly identifies the peak efficiency region around $J = 0.3-0.4$ and maintains the proper slope relationships between thrust, torque, and advance ratio, suggesting it captures the fundamental physics of propeller operation while needing refinement in specific aerodynamic modeling to improve quantitative accuracy.

Conclusions

In summary, the Momentum-Blade Element Theory provides a valuable engineering tool for preliminary propeller analysis, offering approximately 75-80% accuracy in thrust prediction and 60-70% accuracy in torque estimation. The method proves most reliable in the moderate advance ratio range of $J = 0.2-0.5$, where small angle assumptions remain valid and the model captures the essential performance trends. While the current implementation shows limitations in predicting stall behavior and high-advance ratio performance, its computational efficiency and reasonable accuracy make it suitable for initial design studies and trend analysis. The overpredictions observed suggest opportunities for improvement through more sophisticated aerodynamic modeling, particularly by incorporating variable drag coefficients based on Reynolds number effects, adding tip loss corrections, and implementing better stall models. For practical

engineering applications, the MBET approach serves as an effective first-order analysis tool that can guide propeller selection and provide initial performance estimates before proceeding to more detailed computational or experimental validation.

```
clear all; close all; clc;

%% Propeller and Flight Parameters

D = 9 * 0.0254;          % Diameter in meters (9 inches to m)

R = D/2;                  % Radius in meters

RPM = 6000;

Omega = RPM * 2*pi/60; % rad/sec

n = RPM/60;               % rev/sec

rho = 1.225;              % air density kg/m^3

V = 0:1:17;              % forward speed range m/s

B = 2;                    % number of blades

A = pi * R^2;             % Disc area

% Airfoil properties

cd = 0.05;                % constant drag coefficient

alpha_0 = -6 * pi/180; % zero-lift angle in radians

a0 = 5.2;                 % lift curve slope per radian

%% Propeller Geometry Data (APC SF 9x4.7 from UIUC)

r_R = [0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.45, 0.50, 0.55, 0.60, 0.65,
0.70, 0.75, 0.80, 0.85, 0.90, 0.95, 1.00]; % radial stations

chord_R = [0.127, 0.135, 0.158, 0.178, 0.195, 0.209, 0.219, 0.225,
0.227, 0.226, 0.221, 0.212, 0.199, 0.182, 0.161, 0.135, 0.097, 0.058]; %
chord/R

beta_chord = [27.54, 25.28, 26.08, 25.47, 24.07, 22.18, 20.00, 18.18,
16.38, 14.83, 13.63, 12.56, 11.56, 10.65, 9.68, 8.51, 6.72, 4.89] *
pi/180; % chord line twist

% Convert to absolute dimensions

r = r_R * R;

chord = chord_R * R;

% Adjust for zero-lift line (add 6 degrees to geometric twist)

beta = beta_chord + 6 * pi/180;

% Solidity at each station ( $\sigma = B*c/(\pi*R)$ )
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```

sigma = (B * chord) ./ (pi * R);

%% Initialize arrays for results

CT_rotor = zeros(size(V)); % Rotorcraft coefficient:  $C_T = T / (\rho A (\Omega R)^2)$ 
CQ_rotor = zeros(size(V)); % Rotorcraft coefficient:  $C_Q = Q / (\rho A (\Omega R)^2 R)$ 

J = V / (n * D); % Advance ratio

%% Momentum-Blade Element Theory Calculation
for i = 1:length(V)

    CT_total = 0;
    CQ_total = 0;

    lambda_c = V(i) / (Omega * R); %  $\lambda_c = V / (\Omega R)$ 

    for j = 1:length(r)
        x = r_R(j); % Non-dimensional radius

        % Initial guess for induced inflow ratio
        lambda_i = 0.1;

        % Iterative solution for inflow ratio (Equation 35 in lecture)
        max_iter = 100;
        tol = 1e-6;

        for iter = 1:max_iter
            lambda = lambda_c + lambda_i; % Total inflow ratio

            % Equation 35:  $\lambda^2 + (\sigma a / 8 - \lambda_c) \lambda - (\sigma a / 8) \beta x = 0$ 
            % Rearranged to solve for  $\lambda_i$ 
            lambda_new = (sigma(j) * a0 / (8 * lambda)) * (beta(j) * x -
lambda);

            % Check convergence
            if abs(lambda_new - lambda_i) < tol

```

```

        break;

    end

    lambda_i = 0.25 * lambda_new + 0.75 * lambda_i; % relaxation
end

lambda = lambda_c + lambda_i;

% Differential thrust coefficient (Equation 31)

dCT = (sigma(j) * a0 / 2) * (beta(j) * x^2 - lambda * x) *
(r_R(2)-r_R(1));

% Differential torque coefficient (Equation 32)

dCQ = (sigma(j) * a0 / 2 * (beta(j) * x * lambda - lambda^2) +
...
sigma(j) * cd / 2 * x^2) * x * (r_R(2)-r_R(1));

CT_total = CT_total + dCT;
CQ_total = CQ_total + dCQ;
end

CT_rotor(i) = CT_total;
CQ_rotor(i) = CQ_total;
end

%% Convert to fixed wing propeller coefficients for comparison with UIUC
data

% Fixed-wing coefficients:

% CT_f = T/(rho*n^2*D^4), CQ_f = Q/(rho*n^2*D^5)

% Conversion formulas from lecture notes:

CT_fixed = CT_rotor * (pi^3 / 4); % Convert C_Tr to C_Tf
CQ_fixed = CQ_rotor * (pi^3 / 8); % Convert C_Qr to C_Qf
CP_fixed = 2 * pi * CQ_fixed; % C_Pf = 2piC_Qf

%% UIUC Experimental Data

J_uiuc = [0.105, 0.134, 0.160, 0.188, 0.216, 0.243, 0.270, 0.295, 0.321,
0.352, 0.382, 0.404, 0.432, 0.457, 0.481, 0.512, 0.539];

```

```

CT_uiuc = [0.1065, 0.1023, 0.0985, 0.0943, 0.0899, 0.0857, 0.0813,
0.0771, 0.0726, 0.0667, 0.0607, 0.0564, 0.0501, 0.0446, 0.0387, 0.0308,
0.0239];

CP_uiuc = [0.0466, 0.0461, 0.0455, 0.0448, 0.0441, 0.0434, 0.0426,
0.0419, 0.0410, 0.0397, 0.0381, 0.0371, 0.0353, 0.0337, 0.0321, 0.0299,
0.0279];

CQ_uiuc = CP_uiuc / (2*pi);

%% Plot Results - Fixed Wing Coefficients for comparison with UIUC

figure;

% Thrust Coefficient Comparison

subplot(2,1,1);

plot(J, CT_fixed, 'b-', 'LineWidth', 2, 'DisplayName', 'MBET
Prediction');

hold on;

plot(J_uiuc, CT_uiuc, 'ro--', 'LineWidth', 1.5, 'MarkerSize', 6,
'DisplayName', 'UIUC Experimental');

xlabel('Advance Ratio J = V/nD');

ylabel('Thrust Coefficient C_T');

title('APC SF 9x4.7 Propeller - Thrust Coefficient Comparison
(Fixed-Wing Convention)');

legend('Location', 'best');

grid on;

% Torque Coefficient Comparison

subplot(2,1,2);

plot(J, CQ_fixed, 'b-', 'LineWidth', 2, 'DisplayName', 'MBET
Prediction');

hold on;

plot(J_uiuc, CQ_uiuc, 'ro--', 'LineWidth', 1.5, 'MarkerSize', 6,
'DisplayName', 'UIUC Experimental');

xlabel('Advance Ratio J = V/nD');

ylabel('Torque Coefficient C_Q');

title('APC SF 9x4.7 Propeller - Torque Coefficient Comparison
(Fixed-Wing Convention)');

legend('Location', 'best');

grid on;

%% Display numerical results

```

```
fprintf('APC SF 9x4.7 at %d RPM\n', RPM);  
fprintf('J\t\tCT_fixed\tCQ_fixed\n');  
for i = 1:length(J)  
    fprintf('%.2f\t\t%.4f\t\t%.4f\n', J(i), CT_fixed(i), CQ_fixed(i));  
end  
fprintf('\nUIUC Experimental Data:\n');  
fprintf('J_uiuc\t\tCT_uiuc\t\tCQ_uiuc\n');  
for i = 1:length(J_uiuc)  
    fprintf('%.3f\t\t%.4f\t\t%.4f\n', J_uiuc(i), CT_uiuc(i), CQ_uiuc(i));  
end
```

Q2 — Fixed-Wing UAS Performance

Given & Assumptions

- Wing area $S = 60 \text{ m}^2$
- Max lift coefficient $C_{L \max} = 1.35$
- Zero-lift drag $C_{D0} = 0.035$
- Aspect ratio $AR = 10$, Oswald efficiency $\varepsilon = 0.73$
 $\Rightarrow$ Parabolic polar $C_D = C_{D0} + KC_L^2$ with

$$K = \frac{1}{\pi \varepsilon AR} = \frac{1}{\pi \cdot 0.73 \cdot 10} \approx 0.04360.$$

- Weights: empty $W_{\text{empty}} = 45,000 \text{ N}$, max fuel $W_f = 3,000 \text{ N}$
 $\Rightarrow W_0 = 48,000 \text{ N}$ (start), $W_1 = 45,000 \text{ N}$ (end).
- Atmosphere: ISA (0–15 km), sea-level density $\rho_0 = 1.225 \text{ kg/m}^3$.
- ISA density model used (0–20 km)
0–11 km (troposphere, linear lapse):

$$T(h) = T_0 - Lh, \quad p(h) = p_0 \left(\frac{T(h)}{T_0} \right)^{\frac{g_0}{RL}}, \quad \rho(h) = \frac{p(h)}{RT(h)}.$$

11–20 km (isothermal at T_{11}):

$$T(h) = T_{11}, \quad p(h) = p_{11} \exp \left[- \frac{g_0(h-11,000)}{RT_{11}} \right], \quad \rho(h) = \frac{p(h)}{RT(h)}.$$

- **Constants:** $g_0 = 9.80665 \text{ m/s}^2$, $R = 287.05287 \text{ J/(kg}\cdot\text{K)}$,
 $T_0 = 288.15 \text{ K}$, $p_0 = 101,325 \text{ Pa}$, $L = 0.0065 \text{ K/m}$,
 $T_{11} = 216.65 \text{ K}$, $p_{11} = p_0 (T_{11}/T_0)^{g_0/(RL)}$.
- Jet: sea-level available thrust $T_{A,0} = 5,000 \text{ N}$, TSFC $c_t = 0.5 \text{ hr}^{-1}$.
Thrust vs altitude: $T_A(h) = T_{A,0} \rho(h)/\rho_0$.
- Prop: efficiency $\eta = 0.85$, SFC $c_p = 0.05 \text{ N/(W}\cdot\text{hr)}$.
Power available vs altitude: $P_A(h) = P_{A,0} \sqrt{\rho/\rho_0}$.

(a) Jet: Maximum Range at Sea Level

The Closed form formula from the lecture:

$$\left(\frac{C_L^{1/2}}{C_D}\right)_{max} = \frac{3}{4} \left(\frac{1}{3KC_{D0}^3}\right)^{1/4}$$

Numerical substitution:

$$C_{D0}^3 = 0.035^3 = 4.2875 \times 10^{-5},$$

$$3KC_{D0}^3 = 3(0.04360)(4.2875 \times 10^{-5}) \approx 5.607 \times 10^{-6},$$

$$\left(\frac{1}{3KC_{D0}^3}\right)^{1/4} = (1/5.607 \times 10^{-6})^{1/4} \approx 20.56,$$

$$\Rightarrow \left(\frac{C_L^{1/2}}{C_D}\right)_{max} = \frac{3}{4} \times 20.56 = 15.42.$$

Lecture formula:

$$R_{max} = 2\sqrt{\frac{2}{\rho S}} \frac{1}{c_t} \left(\frac{C_L^{1/2}}{C_D}\right)_{max} \left(\sqrt{W_0} - \sqrt{W_1}\right)$$

Unit handling

To get R in meters with SI inputs, take c_t in s^{-1} :

$$c_t = 0.5 \text{ hr}^{-1} = \frac{0.5}{3600} s^{-1} = 1/7200 s^{-1}.$$

Hence $1/c_t = 7200 \text{ s}$.

Numerical substitution

$$2\sqrt{\frac{2}{\rho S}} = 2\sqrt{\frac{2}{(1.225)(60)}} = 2\sqrt{0.02721} = 0.32994,$$

$$\sqrt{W_0} - \sqrt{W_1} = \sqrt{48000} - \sqrt{45000} = 219.089 - 212.132 = 6.957.$$

Now multiply:

$$R_{max} = 0.32994 \times 7200 \times 15.42 \times 6.957 = 2.54685 \times 10^5 \text{ m} = 254.685 \text{ km}.$$

Result

$$R_{max} \approx 2.55 \times 10^2 \text{ km} = 254.685 \text{ km}$$

(b) Prop: Maximum Range at Sea Level

Closed form formula for a propeller aircraft at constant altitude, maximum range occurs

at $C_{D0} = C_{Di} \Rightarrow C_L = \sqrt{C_{D0}/K}$, $C_D = 2C_{D0}$. Hence

$$\left(\frac{C_L}{C_D}\right)_{max} = \frac{1}{2\sqrt{KC_{D0}}}.$$

Plugging into the range expression gives the closed form:

$$R_{max} = \frac{\eta/c_p}{2\sqrt{KC_{D0}}} \ln \frac{W_0}{W_1}$$

To express R in meters when c_p is in $N/(W \cdot hr)$, multiply by 3600 (since $1 \text{ W} \cdot hr = 3600 \text{ N} \cdot m$):

$$R_{max}(m) = \frac{\eta/c_p}{2\sqrt{KC_{D0}}} \ln \frac{W_0}{W_1} \times 3600.$$

Numerical substitution

1. Aerodynamic factor

$$2\sqrt{KC_{D0}} = 2\sqrt{0.04360 \times 0.035} = 2\sqrt{0.001526} = 0.07817$$

$$\Rightarrow \frac{1}{2\sqrt{KC_{D0}}} = 12.8065.$$

2. Propulsion & weight terms

$$\frac{\eta}{c_p} = \frac{0.85}{0.05} = 17.0 \text{ [W} \cdot hr/N], \quad \ln \frac{W_0}{W_1} = \ln \frac{48,000}{45,000} = 0.0645385.$$

3. Range

$$R_{max}(m) = 17.0 \times 12.8065 \times 0.0645385 \times 3600 = 50.553 \text{ km}.$$

Result

$$R_{max, prop, SL} \approx 50.553 \text{ km}$$

(c) Jet max range (ideal, fixed $C_{L,opt}$) vs altitude (ISA)

We use the parabolic polar $C_D = C_{D0} + KC_L^2$. For jets, the **maximum range** at a given altitude occurs at the aerodynamic condition that maximizes $(\sqrt{C_L}/C_D)$; denote this lift coefficient by $C_{L,opt}$. Holding $C_L = C_{L,opt}$ fixed across altitudes, the lecture's closed-form jet range is

$$R_{max} = 2\sqrt{\frac{2}{\rho S} \frac{1}{c_t} \left(\frac{\sqrt{C_L}}{C_D} \right)_{max}} \left(\sqrt{W_0} - \sqrt{W_1} \right).$$

At the chosen optimum, every factor except the air density ρ is a constant; hence $R \propto \rho^{-1/2}$. It is therefore convenient to normalize by the sea-level value and write the altitude dependence as

$$R(h) = R_{SL} \sqrt{\frac{\rho_0}{\rho(h)}},$$

where R_{SL} is the **sea-level maximum range** (computed in part (a) from the same closed-form) and $\rho_0 = 1.225 \text{ kg/m}^3$. The only input varying with altitude is $\rho(h)$, which we obtain from the International Standard Atmosphere (ISA). The maximum range of jet-based design increase with altitude.

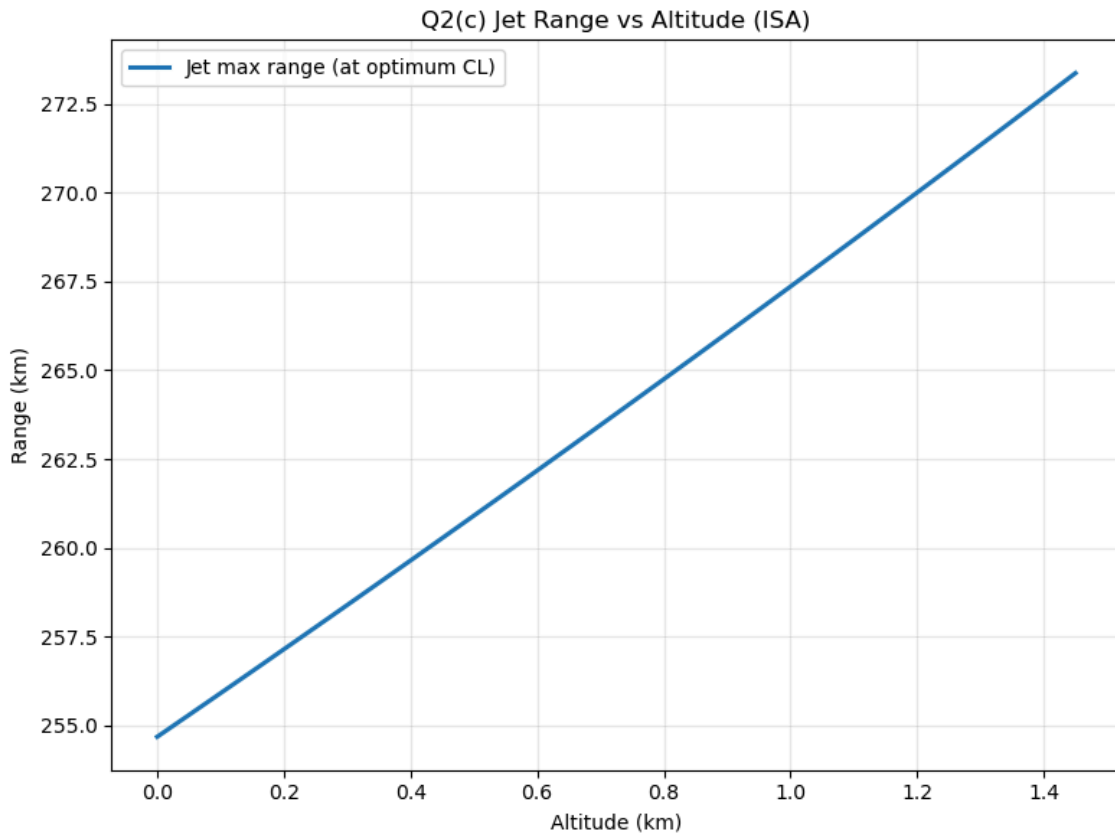


Figure3: Jet-based design range in function of altitude

(d) Maximum Endurance at Sea Level

(1) Jet — Maximum Endurance

Closed-form:

$$E_{max} = \frac{1}{c_t} \left(\frac{C_L}{C_D} \right)_{max} = \frac{1}{2c_t \sqrt{KC_{D0}}} \ln \frac{W_0}{W_1}$$

Substitution

$$\sqrt{KC_{D0}} = \sqrt{0.04360 \cdot 0.035} = 0.03944, \quad \ln \frac{W_0}{W_1} = \ln \frac{48000}{45000} = 0.0645385, \quad \frac{1}{c_t} = \frac{1}{0.5} = 2.0 \text{ hr.}$$

$$\Rightarrow E_{max} = \frac{1}{2 \cdot 0.5 \cdot 0.03944} \times 0.0645385 = 1.652 \text{ hr.}$$

Unit note. With c_t in hr^{-1} , E comes out in *hours*.

(2) Propeller — Maximum Endurance

Closed-form:

$$E_{max} = \frac{\eta}{c_p} \sqrt{2\rho S} \left(\frac{C_L^{3/2}}{C_D} \right)_{max} \left(\frac{1}{\sqrt{W_1}} - \frac{1}{\sqrt{W_0}} \right)$$

At the minimum-power (endurance-optimum) condition for a parabolic polar:

$$C_{D0} = \frac{1}{3} C_{Di} \Rightarrow KC_L^2 = 3C_{D0}, \quad C_{L,PR} = \sqrt{\frac{3C_{D0}}{K}}, \quad C_{D,PR} = 4C_{D0},$$

hence

$$\left(\frac{C_L^{3/2}}{C_D} \right)_{max} = \frac{(\sqrt{3C_{D0}/K})^{3/2}}{4C_{D0}}.$$

Substitution (sea level)

$$\sqrt{2\rho S} = \sqrt{2 \cdot 1.225 \cdot 60} = 12.12436, \quad \left(\frac{1}{\sqrt{W_1}} - \frac{1}{\sqrt{W_0}} \right) = 1.4969 \times 10^{-4},$$

$$\left(\frac{C_L^{3/2}}{C_D} \right)_{max} = \frac{(\sqrt{3 \cdot 0.035 / 0.04360})^{3/2}}{4 \cdot 0.035} = 13.8076, \quad \frac{\eta}{c_p} = \frac{0.85}{0.05} = 17.0.$$

$$\Rightarrow E_{max} = 17.0 \times 12.12436 \times 13.8076 \times 1.4969 \times 10^{-4} = 0.426 \text{ hr.}$$

Feasibility check (stall limit). The endurance-optimum C_L is

$$C_{L,PR} = \sqrt{3C_{D0}/K} = \sqrt{0.105/0.04360} = 1.552 > C_{Lmax} = 1.35.$$

So the theoretical optimum is slightly **below stall speed** (unattainable). If we cap at $C_L = C_{Lmax}$:

$$C_D = C_{D0} + KC_{Lmax}^2 = 0.035 + 0.04360 \cdot 1.35^2 = 0.11447, \quad \frac{C_L^{3/2}}{C_D} = \frac{1.35^{3/2}}{0.11447} = 13.7030,$$

$$\Rightarrow E_{max}^{(C_L=C_{Lmax})} = 17.0 \times 12.12436 \times 13.7030 \times 1.4969 \times 10^{-4} = 0.423 \text{ hr.}$$

Unit note. η/c_p has units $W \cdot \text{hr}/N$; combined with the other factors, E is in *hours*.

(e) Prop Endurance vs Altitude

Endurance formula (propeller, constant altitude)

$$E(h) = \frac{\eta}{c_p} \sqrt{2\rho(h)S} \frac{C_L^{3/2}}{C_D} \left(\frac{1}{\sqrt{W_1}} - \frac{1}{\sqrt{W_0}} \right).$$

Maximum endurance for a parabolic polar occurs at the **minimum-power point**, i.e. when $C_L^{3/2}/C_D$ is maximized:

$$KC_L^2 = 3C_{D0} \Rightarrow C_{L,PR} = \sqrt{\frac{3C_{D0}}{K}}, \quad C_{D,PR} = 4C_{D0}, \quad \left(\frac{C_L^{3/2}}{C_D} \right)_{max} = \frac{(\sqrt{3C_{D0}/K})^{3/2}}{4C_{D0}}.$$

Stall feasibility With the given data,

$$C_{L,PR} = \sqrt{\frac{3 \cdot 0.035}{0.04360}} = 1.552 > C_{Lmax} = 1.35,$$

so the theoretical minimum-power point is **not attainable**. We therefore cap the operating point at $C_L = C_{Lmax}$ and use

$$\left(\frac{C_L^{3/2}}{C_D} \right)_{use} = \frac{C_{Lmax}^{3/2}}{C_{D0} + KC_{Lmax}^2}.$$

Numerically,

$$C_D(C_{Lmax}) = 0.035 + 0.04360 \cdot 1.35^2 = 0.11447, \quad \left(\frac{C_L^{3/2}}{C_D} \right)_{use} = \frac{1.35^{3/2}}{0.11447} = 13.7030.$$

Plotting relation (collect constants) Group all altitude-independent factors into a constant C_E :

$$E_{max}(h) = C_E \sqrt{\rho(h)}, \quad C_E = \frac{\eta}{c_p} \sqrt{2S} \left(\frac{c_L^{3/2}}{C_D} \right)_{use} \left(\frac{1}{\sqrt{W_1}} - \frac{1}{\sqrt{W_0}} \right).$$

With the numbers:

$$\frac{\eta}{c_p} = \frac{0.85}{0.05} = 17.0, \quad \sqrt{2S} = \sqrt{120} = 10.9545, \quad \left(\frac{1}{\sqrt{W_1}} - \frac{1}{\sqrt{W_0}} \right) = 1.4969 \times 10^{-4} N^{-1/2},$$

so

$$C_E \approx 0.3829 \text{ hr}/\sqrt{kg/m^3}, \quad E_{max}(h) \approx 0.3829 \sqrt{\rho(h)} \text{ (hr)}.$$

Because $E_{max}(h) \propto \sqrt{\rho(h)}$, endurance decrease with altitude for a propeller aircraft (opposite to the jet-range trend). The assignment does not specify an engine power-available model $P_A(h)$, so no thrust/power feasibility clipping is required.

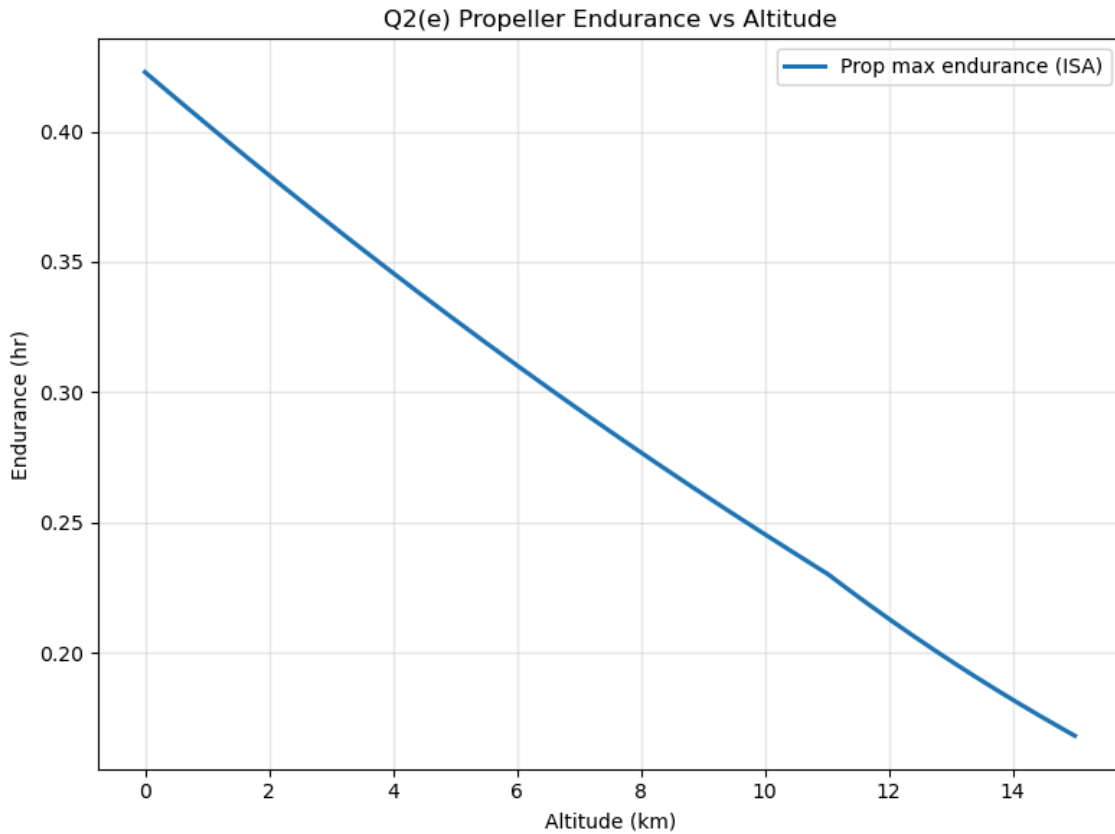


Figure 4: endurance of the prop-based design as a function of altitude over 0–15 km

(f) Velocity–Altitude Envelope up to 15 km

Definitions (three curves in the envelope)

1. Minimum flight (stall) speed.

Defined at the stall limit $C_L = C_{L\max}$.

$$V_{\min,SL} = \sqrt{\frac{2W}{\rho_0 S C_{L\max}}}.$$

2. Minimum thrust-required speed (aerodynamic minimum drag; same condition as $(L/D)_{\max}$).

First compute

$$C_{L,TR} = \sqrt{\frac{C_{D0}}{K}} = \sqrt{\frac{0.035}{0.04360}} = 0.896,$$

then

$$V_{TR,\min,SL} = \sqrt{\frac{2W}{\rho_0 S C_{L,TR}}}.$$

3. Minimum power-required speed (endurance-optimum for props).

First compute

$$C_{L,PR} = \sqrt{\frac{3C_{D0}}{K}} = \sqrt{\frac{0.105}{0.04360}} = 1.552,$$

then

$$V_{PR,\min,SL} = \sqrt{\frac{2W}{\rho_0 S C_{L,PR}}}.$$

Feasibility note: Here $C_{L,PR} = 1.552 > C_{L\max} = 1.35$, so the theoretical minimum-power point is below stall (unattainable). Therefore the realizable minimum-power speed is clamped by stall for all altitudes:

$$V_{PR,\min}(h) = V_{\min}(h).$$

Numerical sea-level values (using $W = 48,000 \text{ N}$, $S = 60$, $\rho_0 = 1.225$):

$$V_{\min,SL} = \sqrt{\frac{96000}{1.225 \cdot 60 \cdot 1.35}} \approx 31.1 \text{ m/s},$$

$$V_{TR,\min,SL} = \sqrt{\frac{96000}{1.225 \cdot 60 \cdot 0.896}} \approx 38.2 \text{ m/s},$$

$$V_{PR,min,SL} = \sqrt{\frac{96000}{1.225 \cdot 60 \cdot 1.552}} \approx 29.0 \text{ m/s } (< V_{min,SL}).$$

Altitude dependence (ISA scaling) For a fixed C_L point, lift balance $W = \frac{1}{2} \rho V^2 S C_L$ implies $V \propto 1/\sqrt{\rho}$. Thus each sea-level speed scales with altitude using ISA density $\rho(h)$:

$$V_{min}(h) = V_{min,SL} \sqrt{\frac{\rho_0}{\rho(h)}},$$

$$V_{TR,min}(h) = V_{TR,min,SL} \sqrt{\frac{\rho_0}{\rho(h)}},$$

$$V_{PR,min}(h) = V_{min}(h) \text{ (for this aircraft).}$$

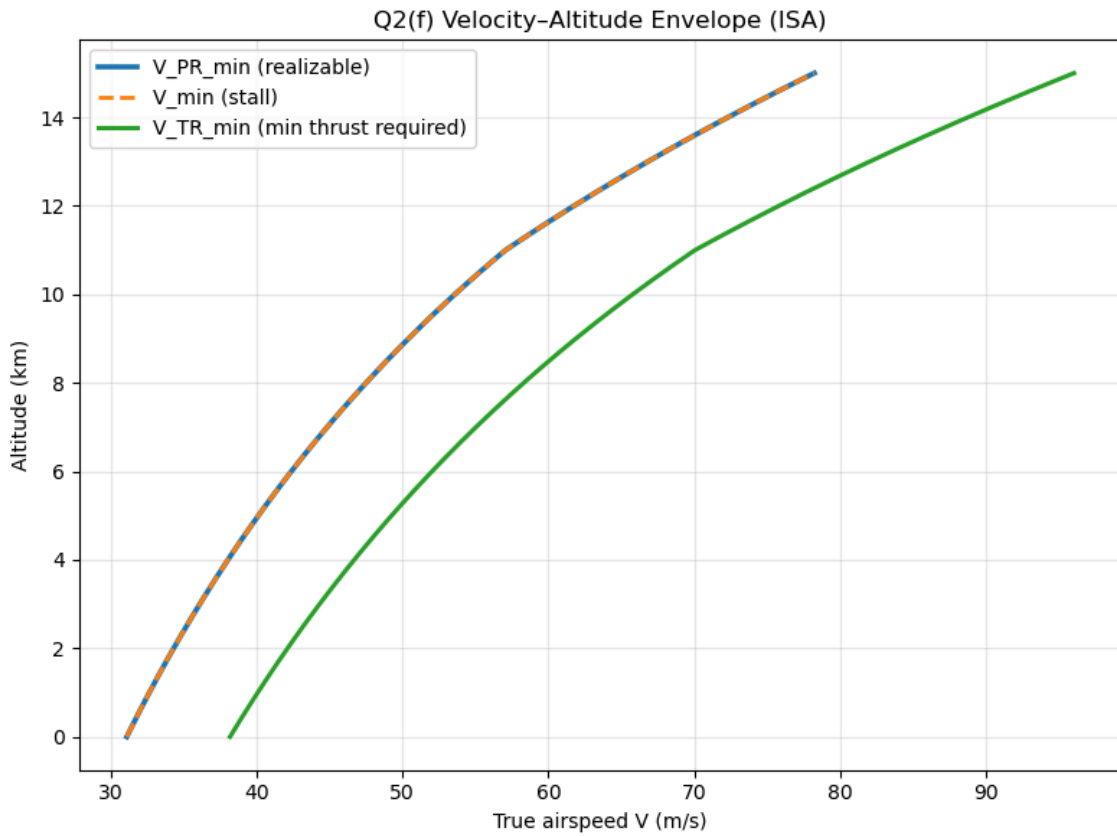


Figure 5: Velocity-Envelope for V_{min} , $V_{TR,min}$ and $V_{PR,min}$

(g) Max Rate-of-Climb vs Altitude and Ceilings

Max Rate of Climb vs Altitude (Jet & Prop, ISA 0–15 km)

Required: For both configurations (jet and propeller), under ISA from 0 to 15 km, compute

$$(R/C)_{max}(h) [m/s]$$

and show it versus altitude h . Indicate:

- Service ceiling: where $(R/C)_{max} = 0.5 \text{ m/s}$,
- Absolute ceiling: where $(R/C)_{max} = 0$.

Jet (thrust-limited)

Climb rate vs speed:

$$\frac{R}{C}(V, h) = V \left[\frac{T}{W} - \frac{1}{2} \rho(h) V^2 \frac{C_{D0}}{(W/S)} - \frac{2K(W/S)}{\rho(h) V^2} \right].$$

Available thrust (lecture simplification):

$$\frac{T}{W}(h) = \frac{T_A(h)}{W} = \frac{T_{A0}}{W} \frac{\rho(h)}{\rho_0}.$$

Speed that maximizes climb rate (the lecture's closed form):

$$V_{RC, max}(h) = \sqrt{\frac{(T/W)(W/S)}{3\rho(h)C_{D0}}} \left[1 + \sqrt{1 + \frac{12C_{D0}K}{(T/W)^2}} \right].$$

Enforce feasibility: $V^* = \max\{V_{RC, max}, V_{min}\}$.

Jet max climb rate:

$$(R/C)_{max, jet}(h) = V^* \left[\frac{T}{W}(h) - \frac{1}{2} \rho(h) V^{*2} \frac{C_{D0}}{(W/S)} - \frac{2K(W/S)}{\rho(h) V^{*2}} \right].$$

Propeller (power-limited)

Lecture closed form:

$$(R/C)_{max, prop}(h) = \frac{P_A(h) - (P_{R, min})(h)}{W}.$$

- If the assignment doesn't specify altitude decay for available propulsive power, take $P_A(h) = P_{A0}$ (constant). If a decay law is specified, use $P_A = P_{A0}(\rho/\rho_0)^\alpha$.
- The **minimum power required** for the parabolic polar comes from $P_R(V) = AV^3 + B/V$ with $A = \frac{1}{2} \rho S C_{D0}$, $B = \frac{2KW^2}{\rho S}$:

$$V_{PR}^* = (B/3A)^{1/4}, \quad (P_R)_{min}(h) = \frac{4}{3^{3/4}} A^{1/4} B^{3/4}.$$

If $V_{PR}^* < V_{min}(h)$ (stall-limited), use

$$(P_R)_{min} = P_R(V_{min}, h) = AV_{min}^3 + B/V_{min}.$$

Result

- Jet (Thrust-limited)
 - Service ceiling ($RC = 0.5 \text{ m/s}$): $h \approx 1488 \text{ m}$
 - Absolute ceiling ($RC = 0$): $h \approx 3000 \text{ m}$
- Prop (Power-limited)
 - Service ceiling ($RC = 0.5 \text{ m/s}$): $h \approx 6563 \text{ m}$
 - Absolute ceiling ($RC = 0$): $h \approx 7750 \text{ m}$

Q3 — Multi Rotor Performance

Given & Assumptions

Aircraft Specifications:

- Total mass: 2 kg (1 kg UAV + 1 kg payload)
- Configuration: Hexarotor (6 rotors)
- Operating altitude: Sea level
- Propellers: APC 8x3.8 slow flyer propellers

Aerodynamic Properties:

- Frame drag coefficient (CD): 0.7
- Reference area (S): 0.01 m^2

Motor Specifications:

- Motor constant (Kv): 1000 rpm/V
- No-load current (i_0): 0.4 A
- Motor resistance (r_m): 0.12 Ω
- Motor torque constant relationship: $K_v K_t = 1$
- Motor efficiency: 70%

Electronic Speed Controller (ESC):

- Internal resistance (r_e): 0.05 Ω
- ESC efficiency: 80%

Battery System:

- Configuration: 3-cell LiPo
- Capacity: 1300 mAh
- Discharge rate: 30C
- Discharge curve coefficients (for discharge in mA-hr):
 - $a = 12.3072$
 - $b = -0.000311$
 - $c = -0.008112$
 - $d = -4.7809 \times 10^{-7}$
 - $e = -7.7835 \times 10^{-7}$
 - $f = 1.4086 \times 10^{-10}$

(a) Hover Power Required

$$\text{Weight} = \text{Mass}_{\text{Total}} \times 9.81 = 19.62\text{N}$$

$$T_{\text{Hover}} = \text{Weight} = 19.62\text{N}$$

$$\text{Diameter} = 8\text{inches} = 0.2032\text{m}$$

$$\text{Sea-level density } \rho_0 = 1.225 \text{ kg/m}^3$$

$$\text{Using actuator disc theory } P_{\text{ind}} = \frac{T^{3/2}}{\sqrt{2\rho A}} \text{ and } A = n \times \pi \times r^2$$

$$P_{\text{ind}} = \frac{19.62^{3/2}}{\sqrt{2(1.225)(6)(\pi)(0.1016)^2}} = 125.87\text{W}$$

As discussed in notes, this is inaccurate, as it does not account for profile losses, Reynold's number effects, etc. We can use data from UIUC to

correct this. The dataset provides rpm, C_t and C_p for our exact propeller (8x3.8) at static conditions. By extrapolating from the UIUC data to find an operating rpm that yields thrust = weight (According to the formula $T = C_t \rho n^2 D^4$). An optimal rpm of 7021 was found with C_T : 0.1143, and C_P : 0.0481. Using the equation

$$P_{Single Rotor} = C_{pf} \rho n^3 D^5 = (0.0481)(1.225 kg/m^3)(7021 rpm \times \frac{1 rpm}{60 rps})^3 (0.2032 m)^5$$

$$= 32.7 \frac{kg \cdot m^2}{s^3} = 32.7 W$$

Therefore $Power_{Hover} = 32.7 \times 6 = \mathbf{196.28W}$

This represents a figure of merit difference of $125.87 / 196.28 = 64.1\%$. This is consistent with the lecture 5 slide 10 analysis of the same propeller at a similar speed (7376RPM yielded $20.7/37.9 = 54.6\%$ accuracy).

See Q3a.m attached for extrapolation script.

This 196.28W represents the total electrical power supplied to the propellers necessary to achieve hover, and does not account for 1. Motor and ESC power efficiencies 2. Power lost due to disc interference.

We can correct this using the supplied efficiencies of $\eta_m = 70\%$ and $\eta_{ESC} = 80\%$ to get $P_{TOTAL} = P_{Hover} / (\eta_m \times \eta_{ESC}) = \mathbf{350.5W}$ that must be supplied by the battery to hover. Disc interference will likely increase this number further.

(b) Hover Endurance

Assumptions:

$$D_0 = 1 \text{ mAh}; \Delta t = 0.1s;$$

Standard voltage batteries @ 3.7V

By implementing the algorithm from lecture 5 slide 20 and using our hover analysis above we can solve the hover endurance time.

Unknowns:

$$\Omega = 7021 \text{ RPM} \times \frac{2\pi}{60} = 735.2 \text{ rad/s}$$

Torque Per Motor (Q) =

$$C_{pf} \rho n^3 D^5 / \Omega = 32.7 W / 735.2 s^{-1} = 0.0445 N \cdot m$$

Motor current (i_m) =

$$\frac{Q}{K_t} + i_0 = (0.0445 Nm)[(1000 RPM/V)(\frac{2\pi}{60s})] + 0.4A = 5.06A$$

Motor voltage (v_m) =

$$\frac{\Omega}{K_v} + i_m r_m = (735.2 rad/s)/[(1000 RPM/V)(\frac{2\pi}{60s})] + (5.06A)(0.12\Omega) = 7.63V$$

$$\text{ESC Voltage } (v_{et}) = v_m + i_m r_e = (7.63) + (5.06)(0.05) = 7.88V$$

Using the given coefficients (**NOTE: Traub uses a different order of coefficients than those given in the notes, as he is the source, his order was used) and solving for g, we can substitute the 3 equations to find

$$k_e = \left(\frac{v_{et} i_m^{0.05}}{g} \right)^{(20/19)} \quad \text{and} \quad i_b = i_m k_e$$

Using the step march defined as $D = D + i_b \Delta t$ and stopping at $k_e > 1$, the battery has a total endurance of 174s for this power rating, discharging 1162.4 mAh.

See attached q3b.m for step march

(c) Forward Flight Analysis

Using the process laid out in lecture 5 slide 29 we can find the requirements as follows

1. Find frame drag

$$D_f = 1/2 \rho S_{ref} C_{Df} V^2 \text{ at } \alpha_D = 0 \text{ This uses the assumption that } \alpha_D \text{ does not affect drag}$$

Given

$$\text{Sea-level density } \rho_0 = 1.225 \text{ kg/m}^3$$

$$S_{ref} = 0.01 \text{ m}^2$$

$$C_{Df} = 0.7$$

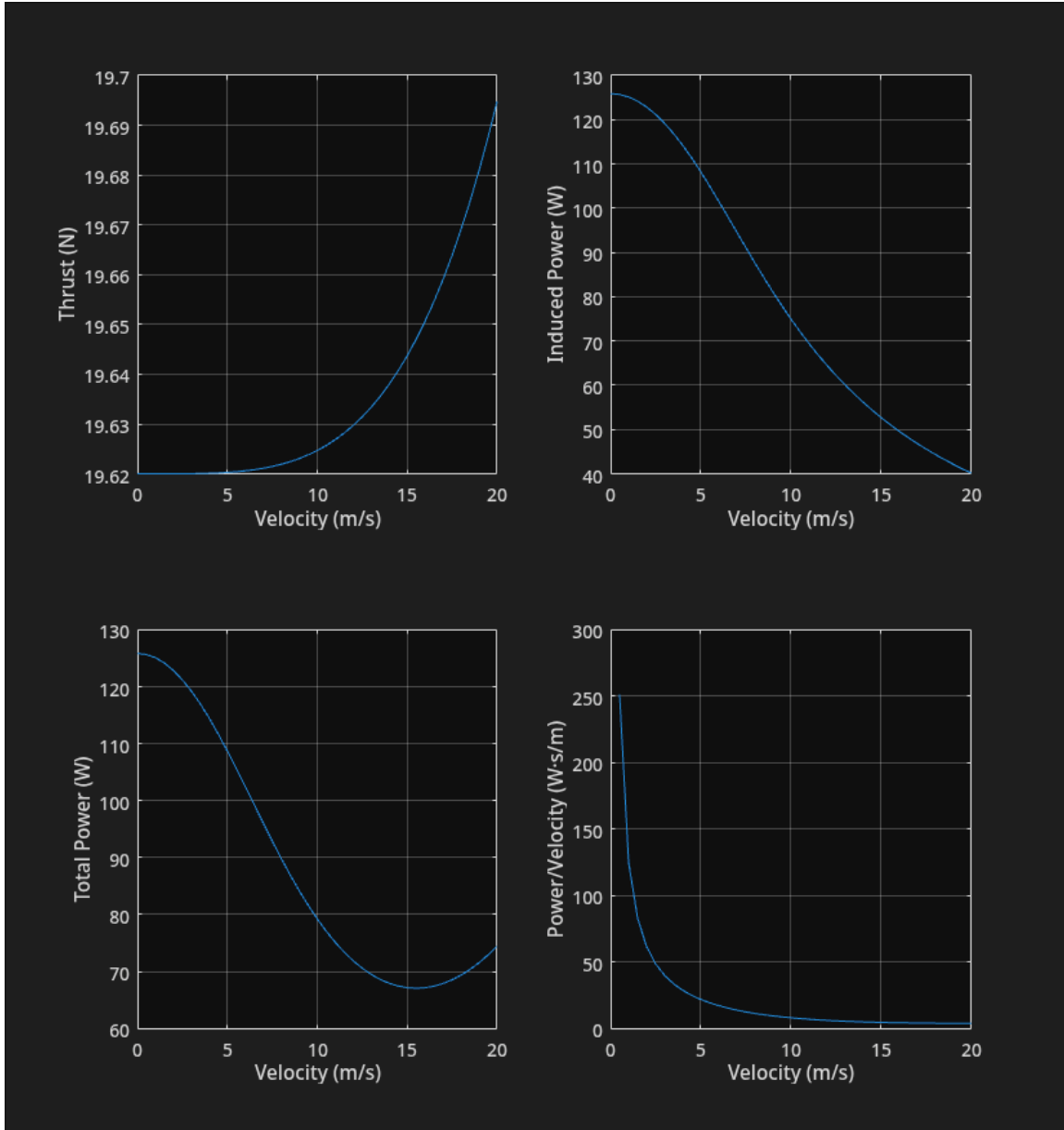
And V is our independent variable.

$$2. \text{ Then we solve for } \alpha_D = \text{atan}(D_f/W) \text{ and Thrust} = \sqrt{W^2 + D_f^2}$$

$$3. \text{ We find the smallest real roots of } v^4 + 2*V(i)\sin(\alpha_D)*v^3 + V(i)^2*v - (W^2 + D_f^2)/(2*\rho*A)^2$$

4. And use this root to solve $P_{\text{ind}} = Tv$ and $P_{\text{Total}} = T(v + V(i)\sin(\alpha_D))$

For improved accuracy, an empirical factor $\kappa \approx 1.15$ could be applied to account for non-uniform inflow effects typical in multirotors.



UIUC Data from <https://m-selig.ae.illinois.edu/props/volume-1/propDB-volume-1.html>